

EXERCISE 5: ANALYSIS OF CUSTOMACO VS MMAS VARIANTS

ALGORITHM DESIGN

CustomACO incorporates several distinctive features:

- 50 ants per generation (vs MMAS single-solution approach)
- Elitism-based pheromone updates using top 20% of ants
- Local search applied with 10% probability (1-bit flip neighbourhood)
- Fixed evaporation rate of 0.2
- Problem specific heuristics slightly favoring 1-bits on OneMax, LeadingOnes, and Linear

PERFORMANCE BY PROBLEM TYPE

Easy Problems (F1 OneMax, F3 Linear): CustomACO performs comparably to all MMAS variants, reaching optimum quickly. The larger population provides no significant advantage on these smooth landscapes where focused exploitation is sufficient.

Sequential Constraint (F2 LeadingOnes): CustomACO performs competitively when compared with MMAS variants. Local search occasionally helps discover beneficial flips, however, the pheromones in the CustomACO struggle to identify and appropriately adapt to the optimal solution effectively.

Deceptive/Difficult Problems (F18 LABS, F23 NQueens, F24 ConcatenatedTrap):

- On LABS, all algorithms struggle equally; CustomACO shows no advantage.
- On NQueens, comparable performance with MMAS; population diversity provides modest benefit.
- On ConcatenatedTrap, CustomACO handles deception reasonably well and elite selection helps prevent premature convergence. MMAS continues to outperform the CustomACO at all stages.

Highly Rugged (F25 NK-Landscapes): Both the CustomACO and MMAS variants struggle immensely in this problem type, as the deep local optima prevents any meaningful learning occurring.

KEY INSIGHTS

Population Size Impact: The 50 ant population increases diversity and provides advantages on deceptive/rugged landscapes but offers limited benefit on smooth problems where MMAS's focused approach is more efficient. In the event a local optima is found the population size offers some minor assistance in resisting the temptation.

Elite Selection: Using top 20% for pheromone updates balances exploitation and exploration better than single solution updates, providing robustness on deceptive problems without sacrificing performance on easy ones.

Local Search: The 10% probability may be too low for consistent effect, and single bit neighborhoods may be insufficient for complex problems. Applying local search allows for valid neighbours of greater quality to be utilised in the creation of a more performative population. Certain local search instances were too computationally expensive (specifically that which occurs within a while better loop).

Fixed Evaporation: The 0.2 evaporation rate provides reasonable balance but may be suboptimal for specific problems. MMAS testing multiple ρ values demonstrates the importance of this parameter.

COMPARATIVE PERFORMANCE

CustomACO does not consistently outperform MMAS variants but demonstrates robust, competitive performance across all benchmarks. While the MMAS variants outperform the CustomACO in the longterm, it proves itself an efficient learner early on in the training phase.

The results confirm that problem characteristics dominate algorithm design choices. On easy problems, the ACO performs incredibly well, but struggles on later harder problems. The optimal configuration is problem dependent rather than universal, and without a more sophisticated dynamic heuristic calculation method, the ACO will continue to struggle.

CONCLUSION

CustomACO represents a well designed population based ACO with competitive performance across diverse problem types. The larger population trades

computational efficiency for robustness providing more consistent results but not faster convergence. Neither aggressive single solution MMAS updates nor population based diversity universally dominates, confirming that algorithm configuration should match problem structure. Future improvements could include adaptive evaporation rates, enhanced local search strategies, and stronger problem specific heuristics.